Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/03/2024
PI Name:	Dunand, David Ph.D.		
Project Title:	Microstructure Evolution in Freeze-Cast Materials		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60208	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	11/06/2017	**End Date:**	03/31/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	Note: End date changed to 03/31/2025 per NSSC information (Ed., 12/3/24). Note: End date changed to 11/05/2024 per NSSC information (Ed., 3/4/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Voorhees, Peter Ph.D. (Northwestern University)		
Grant/Contract No.:	80NSSC18K0196		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Freeze-casting is a novel technique that utilizes ice as a fugitive space holder to fabricate a diverse variety of materials exhibiting elongated, aligned pores. Experimental studies in microgravity inherently simplify the freeze-casting system by minimizing gravity-induced forces that contribute to its complexity, e.g.. sedimentation, buoyancy, and natural convection. Freeze-casting has the potential to produce porous products with specific microstructure including net- and complex-shaped products, provided solidification conditions are properly controlled. Moreover, freeze-casting holds significant promise as an in situ resource utilization technique for space-based materials processing, thus increasing the reliability and safety of access to space while also decreasing overall costs. An improvement in scientific knowledge entails robust and predictive control of materials for a wide variety of applications, thus enabling optimized fabrication on Earth, on planetary surfaces (Moon and Mars), and in orbit.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Freeze-casting has the potential to produce porous products with specific microstructure including net- and complex-shaped products, provided solidification conditions are properly controlled. Moreover, freeze-casting holds significant promise as an in situ resource utilization technique for space-based materials processing, thus increasing the reliability and safety of access to space while also decreasing overall costs. An improvement in scientific knowledge entails robust and predictive control of materials for a wide variety of applications, thus enabling optimized fabrication on Earth, on planetary surfaces (Moon and Mars), and in orbit.
Task Progress:	1. Naphthalene distillation - If air dissolves in liquid naphthalene, air bubble formation is expected during solidification, disrupting the solid-liquid interface. To prevent this issue, lines for transferring liquid suspension from its melting flask to other containers were upgraded from glass to stainless steel. Also, a Pirani pressure gauge and Swagelok connectors are used. A vacuum of 70 mTorr was reached in gas lines after pumping for 40 mins with a roughing pump (Edwards RV8 6.9 CFM). As-received naphthalene (Sigma Aldrich, 99.6% purity) was vacuum distilled prior to use in ampoules. This setup is connected with tubing to a roughing pump to evacuate its air. After reaching a final vacuum of 70 mTorr, pumping was stopped. As-received naphthalene in a glass flask was then melted by heating up to ~92 °C in a silicone oil thermal bath. Evaporated naphthalene was condensed and collected in a glass flask immersed in an ice water bath. Distilled naphthalene is used to prepare suspensions and then filled into assembled ampoules. 2. Design and upgrade of ampoule filling station - In the previous filling, the filling process was interrupted by the unexpected solidification of suspensions to block the access to liquid flow. To solve this issue, the ampoule filling station was upgraded to ensure a homogeneous temperature of the filling setup, i.e., a customized glass funnel attached with an ampoule, above 85 °C (the melting point of naphthalene). Prior to remelting naphthalene suspensions, the glass funnel is connected with recently upgraded stainless steel tubing via a vacuum adaptor to allow evacuation. The distilled naphthalene in the glass flask is remelted in a thermal water bath with a temperature of ~93 °C, equipped with an ultrasonic transducer to prevent sedimentation of particles in suspensions. In step 1, after melting and mixing, the stopcock between the upper glass funnel and ampoule is turned on, enabling the flow of liquid suspension into the ampoule under gravity while sonicating the bath. After filling, hot water in the thermal bath is drained immediately in step 2 under sonification. Immediately thereafter, in step 3, ice water is poured into the bath to solidify the naphthalene rapidly under continuous sonication, to mitigate particle settling and pushing by the solid suspension. 3. Assembled ampoules with a new design - Sample Ampoule Assembly, as designed for the previous Pore Formation Mobility Investigation (PFMI) work, is modified. Redwire is designing the Sample Ampoule Assembly for this work to be similar in construction to those designed previously, components of which include: the main tube body (composed of Schott 8250 borosilicate glass), Kovar piston assembly, cartridge mount head, and six in-situ thermocouples. The inside and outside diameter of the ampoule is ~10.9 and 12.75 mm, respectively. To ensure similar thermal behavior with respect to the glass, Science Requirement 3 specifies an internal diameter tolerance of ±0.2 mm. Based on toxicology assessments conducted by Redwire, the maximum volume of naphthalene that can be used in an individual ampoule is ~6.8 mL. Accordingly, ampoules are filled with ~5.6 mL of science material, corresponding to an effective science material length of ~60 mm, which is much shorter than the overall length of the ampoules (~280 mm, not including the piston assembly or insertion of the Kovar head). After filling in the suspension, an ampoule is evacuated and then sealed with a Kovar piston on the left side to maintain the vacuum of filled suspensions. The spring and Kovar plug are installed. Kovar plugs on both sides are glued to ampoule glass with Torr Seal. 4. Filling of ampoule #2 and #3 - Two ampoules were filled with suspensions: (i) naphthalene and Aerosol-OT (AOT) in Ampoule #2; and (ii) naphthalene, Cu particles (1 μm diameter), and AOT in Ampoule #3, with the upgraded filling station. The filling length in the liquid state is about 60 mm. The first step was to connect the customized glass funnel with an assembled ampoule and then evacuate the whole system to pressures of about 180-200 mTorr. After evacuating, two stopcocks between the glass funnel and the ampoule were switched off to maintain the vacuum level in the ampoule. The upper glass funnel was purged with UHP helium and then loaded with scientific materials. The glass funnel was sealed and then evacuated to vacuum levels of 300-330 mTorr. Meanwhile, the thermal batch was filled with water and heated up to ~93 °C to melt loaded materials. When all naphthalene was melted, the magnetic stirrer was switched on to mix scientific materials (4 h for suspension in Ampoule #2 and 7 h for suspension in Ampoule #3) while sonicating the thermal bath. After mixing the suspensions, the stopcocks were turned on to fill the ampoules with a length of about 6 mm and then turned off immediately to prevent over-filling. The thermal bath was drained out hot water immediately and then refilled with ice water to quench filled suspensions and prevent sedimentation during solidification. 5. Ampoule sealing - After solidification, the filled ampoules were evacuated and sealed with Kovar piston and plug with a customized sealing station provided by Redwire. The ampoule was inserted into a sealing head and then fixed with a hose clamp. The head was then evacuated to a vacuum level of about ~200 mTorr. The Kovar piston was pushed into the ampoule with a steel rod until the upper side of the solidified suspension. After that, a spring and a Kovar plug were installed to seal the ampoule. Kovar plugs on both sides are glued to ampoule glass with Torr Seal.
Bibliography Type:	Description: (Last Updated: 09/30/2024)
Articles in Peer-reviewed Journals	Scotti KL, Voorhees PW, Dunand DC. "Lamellar structures in directionally solidified naphthalene suspensions." Journal of Materials Research. 2024 Aug 14;39(15):2228-35. https://doi.org/10.1557/s43578-024-01381-x , Aug-2024

Journal/Magazine covers	Scotti KL, Voorhees PW, Dunand DC. "Cover in the Journal of Materials Research for article, 'Lamellar structures in directionally solidified naphthalene suspensions.'" Journal of Materials Research. 2024 Aug 14;39(15):2228-35. https://doi.org/10.1557/s43578-024-01381-x , Aug-2024